

CoRoT-2b

R-1471

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_140722 · created 2026-08-01T11:50:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456860.7441 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.016
Transit depth (Rp/Rs)^2	3.62 +/- 0.63 [%]
Orbital Inclination (inc)	89.85 +/- 1.24
Airmass coefficient 1 (a1)	0.868 +/- 0.019
Airmass coefficient 2 (a2)	0.124 +/- 0.063
Scatter in the residuals of the lightcurve fit is	2.25 %
Best Comparison Star	#3 - [264, 253]
Optimal Aperture	3.3
Optimal Annulus	14.61
Transit Duration (day)	0.093 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.016 vs 0.1667 (deviation 1.36σ)
Duration fit vs published	0.093 d vs 0.095 d (deviation 1.04σ)
Mid-time O–C	-4.1 min
Depth SNR	11.1
Residual scatter	2.25 %

Light curve

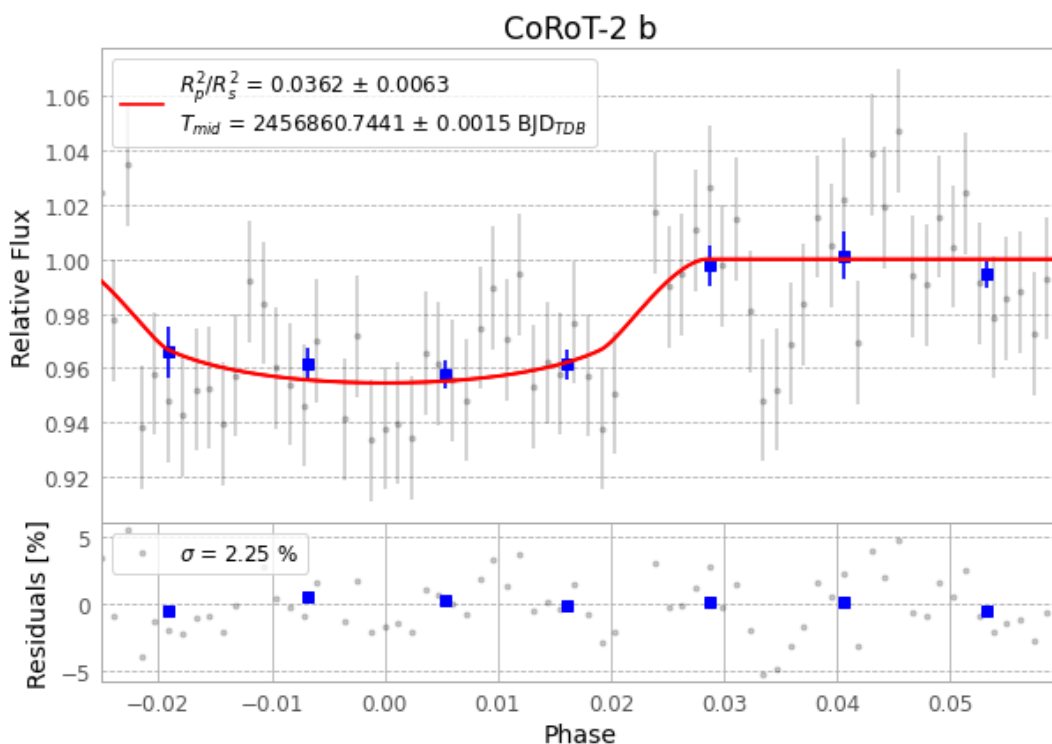


Figure 1 — FinalLightCurve_CoRoT-2 b_2014-07-21.png (SHA-256 a9c2e7e7286b2492...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6219ccb15203e338...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), CoRoT-2140722043913.FITS → CoRoT-2140722081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-140722.log (95daf1854bfc...), FinalLightCurve_CoRoT-2 b_2014-07-21.png (a9c2e7e7286b...), FinalLightCurve_CoRoT-2 b_2014-07-21.csv (ede2dca2c41c...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:50Z.